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Theoretical Atomic Physics and Synthetic Quantum Systems

Kinetic constraints in a Bose-Fermi-Hubbard model

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1 Project motivation

Ongoing research

- Ergodicity and its violation in closed quantum many-body systems is a central open question in modern condensed matter physics.
- Kinetic constraints can induce Hilbert space fragmentation [1], leading to the coexistence of thermalizing and non-thermalizing sectors within the same model.
- Ultracold Bose-Fermi mixtures in optical lattices [2] provide a promising experimental platform for realizing and probing such constrained dynamics.

In this work

→ We propose a Bose-Fermi-Hubbard model on a 1D optical lattice in which inter-species interactions give rise to kinetic constraints on the dynamics of both species.

→ We study the model's non-equilibrium dynamics and thermalization properties in a certain parameter regime.

2 The system

- Introduce (fermionic) bosonic annihilation operators (c_j) b_j obeying canonical (anti-)commutation algebra in 2nd quantization language.
- A staggered potential [3] acting upon the bosons is added to the model.
- For non-interacting bosons the Bose-Fermi-Hubbard Hamiltonian for a 1D chain reads

$$H = \underbrace{-J_B \sum_j (b_j^\dagger b_{j+1} + h.c.) - J_F \sum_j (c_j^\dagger c_{j+1} + h.c.)}_{=:V} + \underbrace{\frac{T}{2}\sum_j (-1)^j b_j^\dagger b_j + U_{BF} \sum_j b_j^\dagger b_j c_j^\dagger c_j}_{=:H_0}, \tag{1}$$

J_B = boson hopping amplitude, J_F = fermion hopping amplitude, T = onsite potential, U_{BF} = boson-fermion interaction strength, U_{BB} = boson-boson interaction strength

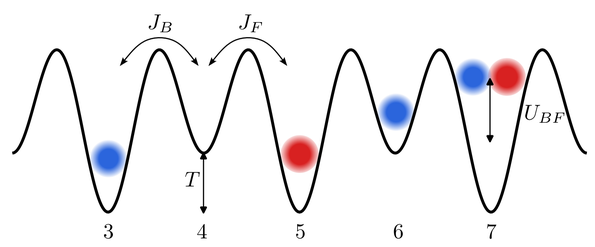

Figure 1: **Schematic depiction of the system.**
Bosons (blue) and fermions (red) in a 1D lattice. The bosonic (fermionic) coherent tunneling at strength J_B (J_F) is described by the operator V in (eq. 2). The bosons are subject to a staggered potential of strength T which enhances (lowers) their energy at even (odd) sites and interact with the fermions with strength U_{BF}, which is described by H_0 in (eq. 2).

3 Effective Hamiltonian

1. Consider the limit $J_B, J_F \ll U_{BF}, T \Rightarrow V$ can be viewed as a perturbation to H_0.
2. Transform V into interaction picture via $U = e^{-iH_0 t}$
$$V^I(t) = U^\dagger V U.$$
3. Tune onsite potential T on resonance with the boson-fermion interaction U_{BF}: $T = U_{BF}$.
4. Apply a rotating-wave approximation (RWA), i. e. keep only time-independent terms in $V^I(t)$.

Effective Hamiltonian

$$H_{eff} = -J_B \sum_{j \text{ odd}} (b_j^\dagger b_{j+1} \overline{\mathcal{P}_{j+1}^F} + b_j^\dagger b_{j-1} \overline{\mathcal{P}_{j-1}^F} + h.c.)\mathcal{P}_j^F - J_F \sum_j (c_j^\dagger c_{j+1} + h.c.)\mathcal{P}_{j+1=j}^B \tag{2}$$

→ $\mathcal{P}_j^F = c_j^\dagger c_j$ is the projector onto the subspace of Fock states containng a fermion at site j and $\overline{\bullet}$ denotes the projector onto the complement

→ $\mathcal{P}_{j+1=j}^B = \sum_{n=0}^\infty |n\rangle\langle n|_j \otimes |n\rangle\langle n|_{j+1}$ is the projector onto the subspace of Fock states of equal boson occupation number at site j and $j+1$

Fermion hopping under kinetic constraint

- In (fig. 2) a (b) fermion hopping between sites j and $j-1$ is allowed as both sites do (not) contain a boson
- In general, hopping is allowed if both sites contain the same amount of bosons, see $\mathcal{P}_{j+1=j}^B$ in (eq. 2).

Boson hopping under kinetic constraint

- In the fermion configurations of (fig. 2) c (d) boson hopping between an odd site k and its left (right) neighbor is allowed
- Those hopping processes are mathematically described by the terms J_B in (eq. 2).

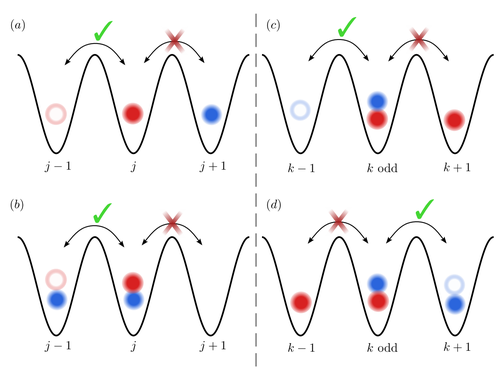

Figure 2: **Elementary processes described by H_{eff}.**

4 One fermion in a bosonic background

- Consider a light fermion immersed in background of heavy bosons, $J_B \ll J_F$.

1. Derive Heisenberg equations of motion (EOM) for the bosonic ladder operators from (eq. 2).
2. Apply *Born-Oppenheimer* approximation (BO) using the seperation of time scales $J_B \ll J_F$.
3. Take expectation values over Heisenberg EOM using BO to decouple fermionic and bosonic DOFs $\langle ... \rangle_{B+F} \approx \langle ... \rangle_B \langle ... \rangle_F$.
4. Apply *Mean-field* approximation (MF) : $|\Psi\rangle_B \approx |\alpha\rangle_1 \otimes |\alpha\rangle_2 \otimes ... \otimes |\alpha\rangle_L \Rightarrow \langle b_j^\dagger b_{j+1}\rangle \approx \langle b_j\rangle^* \langle b_{j+1}\rangle$ with coherent states $|\alpha_j\rangle$ and mean occupation $n_j = |\alpha_j|^2$.

System of coupled equations

MF equations of motion for bosonic amplitudes

$$i\frac{d}{dt} b_j = -J_B \begin{cases} (b_{j+1} + b_{j-1}) \left|\langle j|\Psi\rangle_F\right|^2 & j \text{ odd}, \\ b_{j+1} \left|\langle j+1|\Psi\rangle_F\right|^2 + b_{j-1} \left|\langle j-1|\Psi\rangle_F\right|^2 & j \text{ even} \end{cases} \tag{3}$$
$$- J_F \left(\left(\langle \Psi|j\rangle \langle j+1|\Psi\rangle_F + c.c.\right) \langle B_1\rangle_B + \left(\langle \Psi|j-1\rangle \langle j|\Psi\rangle_F + c.c.\right) \langle B_2\rangle_B \right).$$

Schrödinger equation for the fermionic one-particle wave function $|\Psi\rangle_F$ evolving under H_{eff}^F

$$i\frac{d}{dt}|\psi\rangle_F = H_{eff}^F |\psi\rangle_F, \; H_{eff}^F = -J_B \sum_{j \text{ odd}} \left(|b_j^* b_{j+1}|^2 + |b_j^* b_{j-1}|^2\right) |j\rangle\langle j|_F - J_F \sum_j \langle \mathcal{P}_{j+1=j}^B\rangle_B (|j\rangle\langle j+1|_F + h.c.). \tag{4}$$

5 2 particles on a ring lattice - dynamically disconnected subspaces

- Consider a ring lattice with L sites (L even) and periodic boundary conditions occupied by 1 fermion and 1 boson → Hilbertspace $\mathcal{H} = \text{span}\{|i\rangle_B \otimes |j\rangle_F = |ij\rangle, \; i,j = 1, ...L\}$, dim($\mathcal{H}$) $= L^2$.
- Hilbert space decomposes into $L+1$ dds due to the kinetic constraint, see blue squares in (fig. 3)

→ $L/2$ frozen states $\{|jj\rangle\}$, j even ↔ both particles immobile

→ $L/2$ subspaces span$\{|ji\rangle\}$, j even, $i \neq j$ with dim$=L-1$ ↔ boson immobile, fermion mobile

→ 1 subspace $\mathcal{D}$ ↔ both particles move under kinetic constraint, see the large square in (fig. 3)

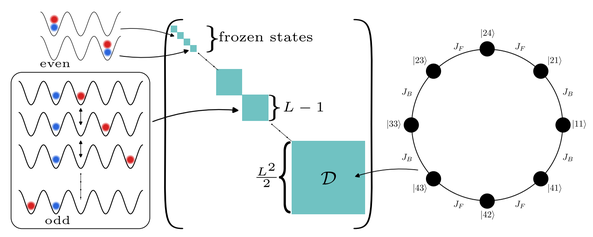

Figure 3: **Structure of the dynamically disconnected subspaces.** The ring structure on the right represents the coupling between Fock states of the projection of H_{eff} onto subspace $\mathcal{D}$, $H_{eff}|_{\mathcal{D}} = \mathcal{P}_D H_{eff} \mathcal{P}_D$ for $L = 4$.

6 Conclusion and outlook

→ in the limit of a light fermion immersed in a background of non-interacting heavy bosons, we derived a MF description using BO approximation

→ dynamics dependent on initial state (ballistic ↔ immobile), effective background potential caused by bosons leads to fermion distribution over sublattice

→ fermion movement suppressed for high boson densities

→ extend MF analysis to a 2D lattice

→ perform systematic analysis of the spectrum of H_{eff} based on the observation of the dds for 2 particles → candidate for HSF?

→ consider hardcore boson limit $U_{BB} \to \infty$

References

[1] M. Will, R. Moessner, and F. Pollmann, "Realization of hilbert space fragmentation and fracton dynamics in two dimensions", Phys. Rev. Lett. **133**, 196301 (2024).

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[3] S. Karch, S. Bandyopadhyay, Z.-H. Sun, A. Impertro, S. Huh, I. P. Rodríguez, J. F. Wienand, W. Ketterle, M. Heyl, A. Polkovnikov, I. Bloch, and M. Aidelsburger, *Probing quantum many-body dynamics using subsystem loschmidt echos*, 2025.

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